

From the 'Net – 4-bit Forth Tom Zimmer

Tom Zimmer posted this item about 4-bit Forth applications on the comp.lang.forth newsgroup after someone described Forth as "requiring a minimum of 16 bits". The very first microprocessor was a 4-bit device – the Intel 4004.

"I have told this story before, but what the hay, one more time;

I was hired at Samsung about 10 years ago, to revamp a micro-controller development system for their line of Super8-based 8-bit processors, and for their line of 4-bit processors (yes, 4-bit) called the 56000. I worked with Jerry Boutel, Robert Smith and others to create a development system that ran on F-PC, and used the customized versions of the TCOM compiler. The target and development system for the Super8 processor was a pretty normal call-threaded implementation, with optimizers that produced code which could be debugged remotely at the source level and was eventually burned into onboard MASK ROM. These applications were limited to 256 bytes of RAM, all that was available on the Super8.

The really interesting target though was the 56000 4-bit processor, which had versions with up to, I think, 32k of ROM, and 256 nibbles⁴ of onboard RAM. Fortunately, this processor had a couple of interesting features. First it had multiple address spaces, one for RAM, one for ROM and several for I/O, so short addressing could be used for all access to RAM, and I/O. Another interesting feature, was the ability to assign byte tokens to subroutines. The TCOM compiler for the 56000 would automatically assign about 16 of the available 48 tokens to predefined Forth primitives, and left the remaining 32 tokens for assignment to user Forth functions.

"4-bit processors are still viable and available from at least a dozen major suppliers. There will always be high volume products (at least 1 million pieces) that need to do some limited processing at very low cost (under \$1). Examples are microwave ovens, bathroom scales and telephones.

We get spoiled with our sub-GHz computers, but the real world doesn't need that kind of processing power for a lot of things. The biggest problem with 4-bit processor is that they tend to be a bear to program. Forth made it more than bearable, it made it fun."

Tom Zimmer 4-Sep-2001

⁴ A "nibble" is 4 bits of data or half a byte. I would love to know who thought of that one – Ed.

The result was that you could generate some truly small programs on the 56000. We added nibble memory operations like N@, and some byte operations like B@. Developing an application truly had the flavor of Forth, even though you were obviously developing for a very specific target processor. The implementation was essentially Forth 83, though portability doesn't really come into play in this environment. If I sound excited, after all these years, I confess I am. It was one of the most interesting development projects I have ever worked on.

Of course Samsung thought everyone would want to program these processors in assembly language, so Robert Smith wrote PASM, a Programmable ASseMbler, which could easily be tailored to any of the processors Samsung built. We did, I believe, have several people that developed their applications in Forth. By the way, the cost of these processors was under a US dollar.

Just my ramblings,

Tom Zimmer"

In a recent poll at www.embedded.com received 1346 responses and listed programming languages for embedded work as:

C	68%
C++	17%
Java	5%
Forth	3%
Other	3%
Ada	2%

his shows that Forth continues to hold its position as a tool for embedded work whose worth is recognised and valued by a minority. (Although you do have to wonder where the assembly-language programmers are. Perhaps they're hidden in the Other category.)